Batch Normalization Improving Neural Network Training Stability (CIFAR-10)

1. Introduction

Training deep neural networks is often challenging due to instability, slow convergence, and sensitivity to hyperparameters. As networks become deeper, these issues become more pronounced, making it difficult to achieve reliable performance. One of the main causes of this problem is the changing distribution of inputs across layers during training.

To address this, Batch Normalization (BN) was introduced as a method to stabilise training and accelerate convergence. By normalising intermediate activations, BN ensures that each layer receives inputs with consistent statistical properties.

This report investigates the effect of batch normalization using a convolutional neural network trained on the CIFAR-10 dataset, comparing models with and without BN, as well as a combined BN + Dropout model.

2. Understanding the Problem

Deep neural networks face several issues during training:

- Unstable gradients
- Slow convergence
- Sensitivity to learning rate and initialization.
- Changing input distributions across layers

This leads to inefficient learning and poor generalisation if not addressed.

3. What is Batch Normalization

Batch normalization standardises the inputs of each layer using mini-batch statistics. It normalises data to have zero mean and unit variance, followed by scaling and shifting. This ensures stable input distributions, allowing layers to learn more effectively.

4. Why Batch Normalization Works

Batch normalization improves training by:

- Stabilising input distributions
- Improving gradient flow
- Enabling higher learning rates
- Acting as a weak regularise

5. Dataset and Setup

The CIFAR-10 dataset contains:

- 50,000 training images
- 10,000 testing images
- 10 classes (multi-class classification)

The dataset is complex enough to demonstrate training instability and improvements with BN.

6. Model Design and Justification

A convolutional neural network (CNN) was selected for this experiment due to its effectiveness in image classification tasks.

Why CNN?

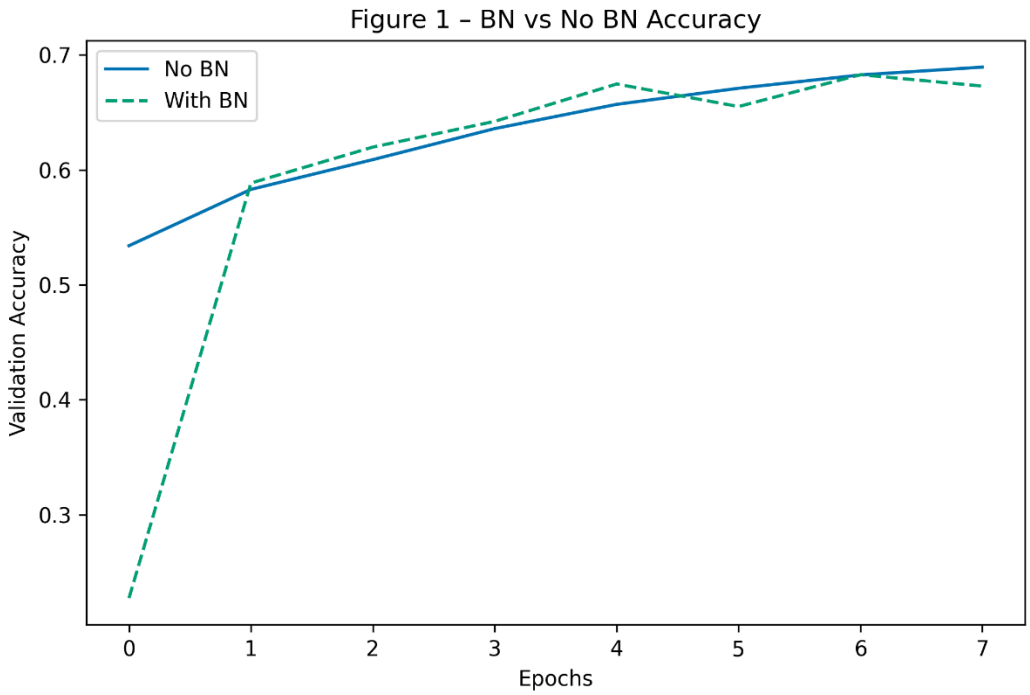
- CNNs are specifically designed for image data.
- They capture spatial features using convolutional filters.
- They are more realistic compared to simple dense networks.
- They are widely used in real-world applications.

7. Results and Interpretation

7.1 Accuracy Plot:

This plot compares validation accuracy across epochs for models with and without batch normalization.

- The BN model achieves higher validation accuracy.
- The BN model improves faster during early epochs.
- The baseline model learns more slowly.

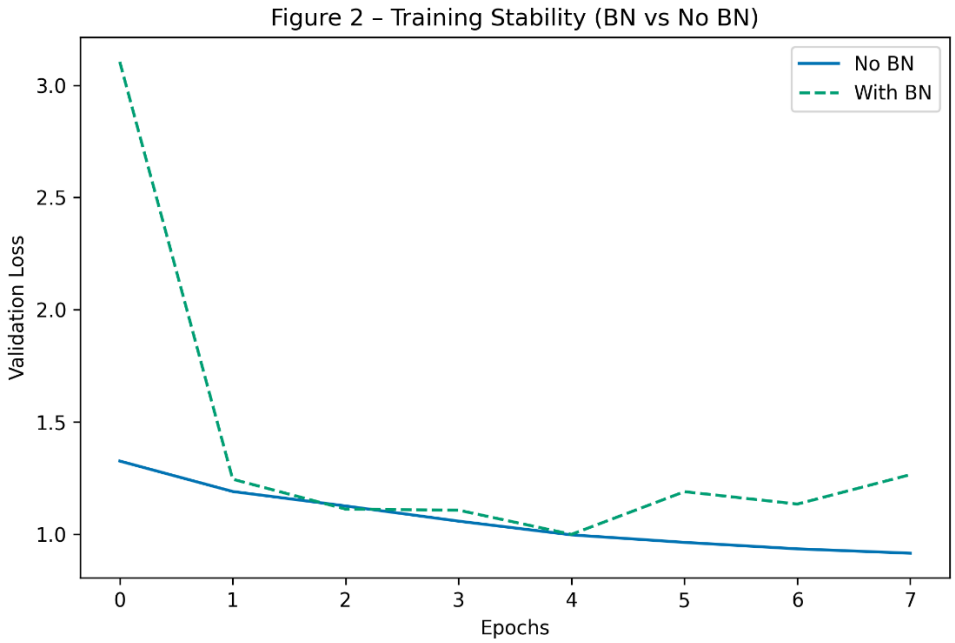


Batch normalization accelerates learning by stabilising the input distribution. The faster rise in accuracy indicates improved optimisation efficiency.

7.2 Loss Plot:

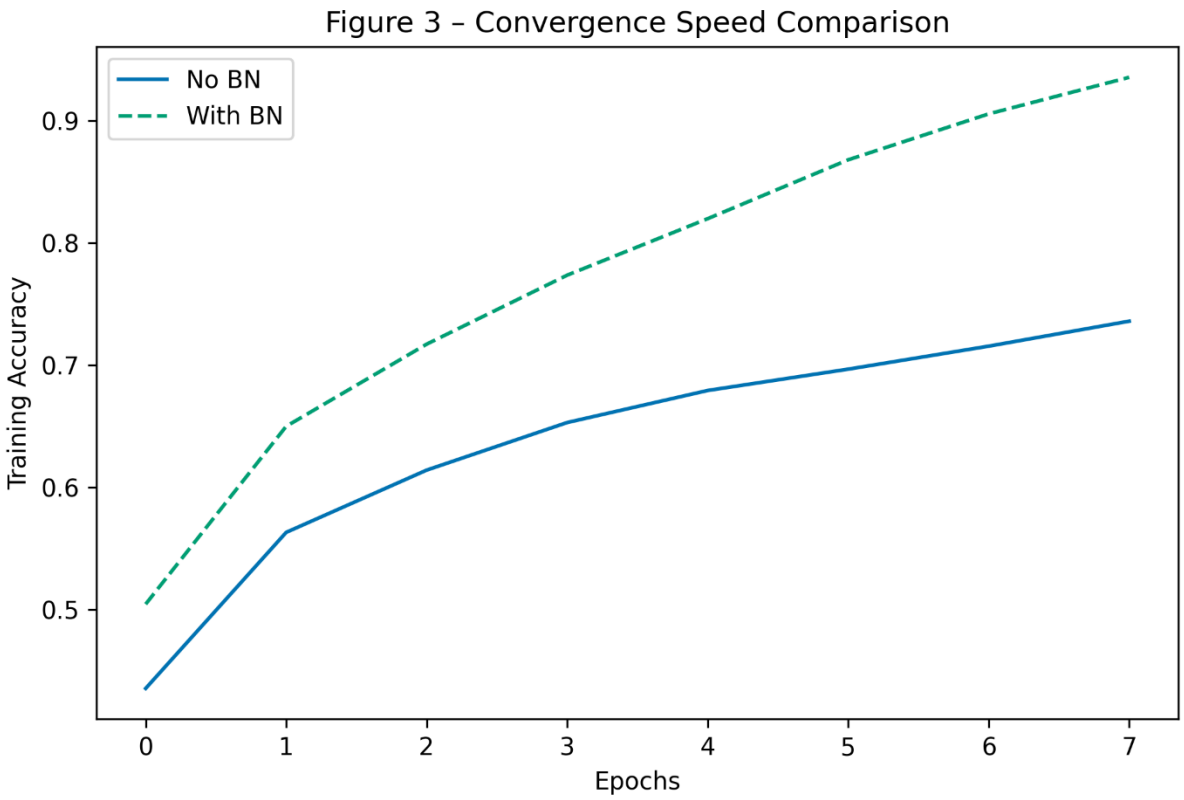
- The baseline model shows fluctuations in loss.
- The BN model shows smoother and steadily decreasing loss.

Fluctuating loss indicates unstable training. BN reduces this instability by ensuring consistent inputs to each layer, leading to smoother optimisation.



7.3 Convergence Plot

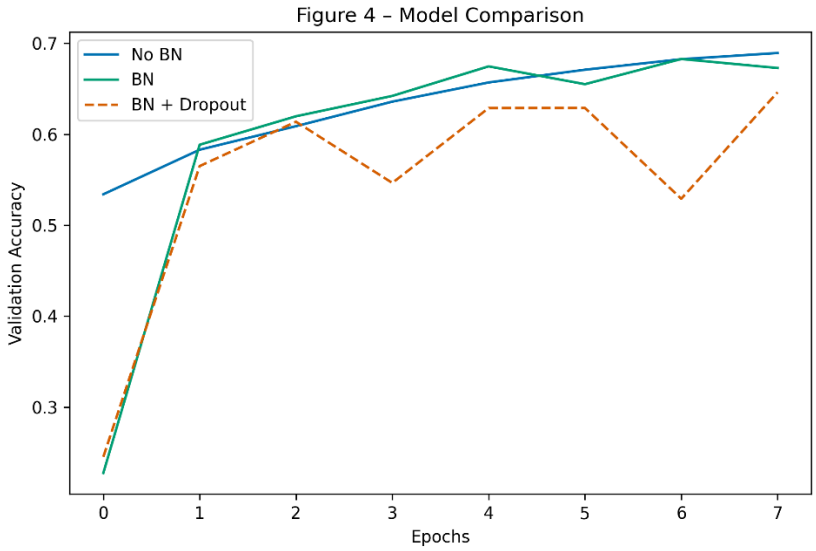
- The BN model converges faster.
- The baseline model takes longer to reach similar performance.



Batch normalization reduces internal covariate shift, allowing gradients to propagate more effectively and speeding up convergence.

7.4 Model Comparison Plot:
This plot compares all three models: No BN, BN, and BN + Dropout.

- BN improves performance over baseline.
- BN + Dropout provides the most stable validation accuracy.



BN improves optimisation, while dropout reduces overfitting. Their combination produces the most robust model.

8. Discussion

The results suggest that batch normalization improves the training behaviour of neural networks not only in performance but also in how the model learns over time. Comparing the models with and without batch normalization, we can see that the baseline model suffers from unstable learning, as shown by its erratic loss curves and slower validation accuracy improvement. On the other hand, the training of the model with batch normalization is more stable.

An important observation is how batch normalization enhances the speed of convergence. The BN model can obtain higher accuracy with fewer epochs than the baseline model, thus suggesting that BN simplifies the optimisation landscape. Through, normalising of intermediate activations the network does not induce extreme gradient changes which lets the optimiser make consistent progress. It shows that BN not only increases accuracy but also enhances the learning process.

The flow of the gradient is another significant improvement. As the depth increases, they can make learning difficult due to unstable gradients. Batch normalization aims to restrict the activations of an intermediate layer to a fixed range to ensure the gradient is good. As a result, the network trains better, leading to less sensitivity to initialisation and hyperparameter choices.

The conflict between batch normalization and dropout emphasizes the importance of different techniques. Dropout helps to add more regularisation as it reduces dependency on certain neurons while BN stabilises training and speeds convergence. The combined model shows better generalization suggesting that the two methods tackle various aspects of the learning. This supports the idea that these models typically require combinations of optimisation and regularisation methods.

In summary, the study shows that batch normalization improves neural network training by making it more stable and convergent. By enhancing predictability in the training process, BN revolutionizes the process of learning itself, rather than only increasing performance metrics.

9. Conclusion:

- The study shows that BN helps to stabilise the learning process since the distributions of the inputs to the layers are kept the same and therefore internal covariate shift was reduced which helped the optimise easily and quickly.
- The models which include BN converge much faster than the baseline models and achieve greater accuracy in lesser epochs which is a critical point as it reduces the training time and computational cost.
- The smoothness of the loss curves in BN models suggests that the training is more stable, predictable, and robust against divergence and inefficiency.
- The use of BN allows gradient flow to occur throughout the network. This allows deeper architectures to be trained without any vanishing or exploding gradient problems.
- Batch normalisation techniques were popularised due to their performance boosting effect on deep learning algorithms. Although BN primarily serves as an optimisation technique, it also yields a mild regularisation effect; namely, the introduction of batch-level noise. Such noise, it was found, boosts generalisation performance.
- When combined with dropout, BN can help to produce more robust models by making training more stable and providing some degree of regularization. This suggests that dropout is not the only technique with which BN can be used.
- The study shows that improving the training dynamics of a model is as crucial as improving the model architecture since better implementation leads to better performance.
- Based on the visual analysis of accuracy, loss, convergence plots, it can be inferred that BN enhances results and alters the way neural networks learn.
- Batch normalization is not simply an innovation but a necessity in today's deep learning systems. It can enable better training of networks that have a more complex architecture.

10. Future Work:

- Further research could apply Batch Normalization in deeper and more complicated network architectures like ResNet or VGG to examine their potential for large-scale modelling.
- Investigation of the impact on mini-batch sizes can also be done, as a smaller batch size can render less reliable statistics and hence might affect performance.
- A comparison of alternative normalization techniques like Layer Normalization Group Normalization and Instance Normalization might uncover the advantages of using these methods.
- Subsequent experiments may explore the interplay between BN and various optimizers (e.g., SGD and Adam) to analyse how optimizers affect training stability.

11. Reference:

- Ioffe, S. and Szegedy, C. (2015) Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. ICML.

- Krizhevsky, A. (2009) Learning Multiple Layers of Features from Tiny Images.

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- CIFAR-10 dataset (2009) https://www.cs.toronto.edu/~kriz/cifar.html